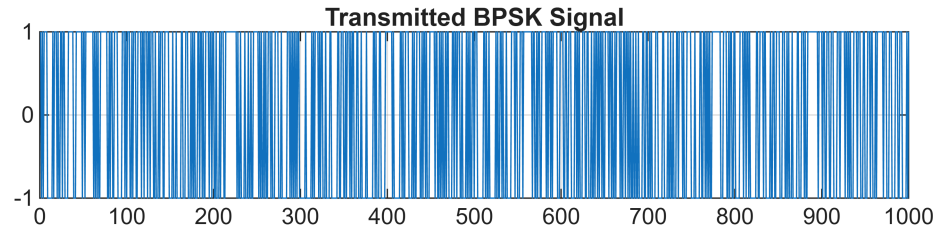
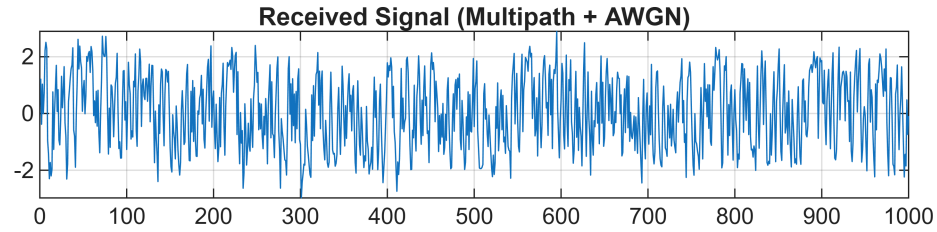
```
clc;
clear;
close all;
% Parameters
N = 1000;
bits = randi([0 1],N,1);
% BPSK Modulation
tx = 2*bits - 1;
% Multipath Fading Channel
h = [1 0.5 0.3];
fade = filter(h,1,tx);
% Add AWGN

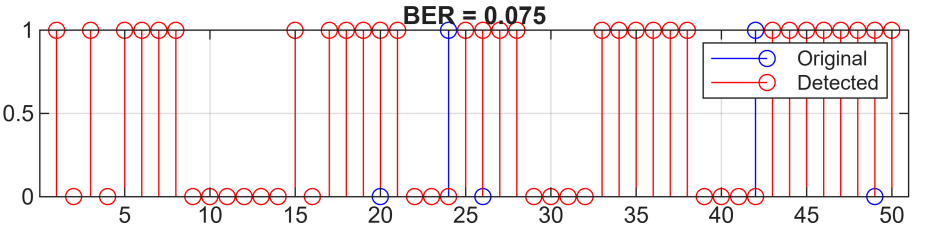

rx = awgn(fade,10,'measured');
% Detection
bits_rx = rx > 0;
% BER Calculation
[~, BER] = biterr(bits,bits_rx);
fprintf('BER = %f\n',BER);
```

```
BER = 0.075000
```

```
% Plots
figure;
subplot(3,1,1)
plot(tx)
title('Transmitted BPSK Signal')
grid on
subplot(3,1,2)
plot(rx)
title('Received Signal (Multipath + AWGN)')
grid on
subplot(3,1,3)
stem(bits(1:50),'b')
hold on
stem(bits_rx(1:50),'r')
title(['BER = ' num2str(BER)])
legend('Original','Detected')
grid on
```


Transmitted BPSK Signal
1
0
-1
0
100
200
300
400
500
600
700
800
900
1000


Received Signal (Multipath + AWGN)
2
0
-2
0
100
200
300
400
500
600
700
800
900
1000


BER = 0.075
1
0.5
0
5
10
15
20
25
30
35
40
45
50
Original
Detected